

Introduction to Chemistry

Section 1.1 The Stories of Two Chemicals

In your textbook, read about the ozone layer.

Use each of the terms below just once to complete the passage.

atmosphere

oxygen gas

ozone

ozone hole

stratosphere

troposphere

ultraviolet radiation

Earth's **(1)** _____ is made up of several layers. The air we breathe makes up the lowest level. This layer is called the **(2)** _____. The next layer up is called the **(3)** _____. This level contains a protective **(4)** _____ layer.

Ozone forms when **(5)** _____ is struck by ultraviolet radiation in the upper part of the stratosphere. The ozone forms a layer around Earth, which absorbs **(6)** _____. Without ozone, you are more likely to get a sunburn or possibly skin cancer. The thinning of the ozone layer, called the **(7)** _____, is worrisome because without ozone all organisms on Earth are subject to harm from too much radiation.

In your textbook, read about chlorofluorocarbons.

For each statement below, write *true* or *false*.

- _____ **8.** CFC is another name for a chlorofluorocarbon.
- _____ **9.** CFCs are made up of carbon, fluorine, and cesium.
- _____ **10.** All CFCs are synthetic chemicals.
- _____ **11.** CFCs usually react readily with other chemicals.
- _____ **12.** CFCs were developed as replacements for toxic refrigerants.

Section 1.2 Chemistry and Matter

In your textbook, read about chemistry and matter.

Define each term.

1. chemistry

2. matter

3. mass

Write each term below under the correct heading. Use each term only once.

air	magnetic field	car	feeling	heat	human body
light	radio	radio wave	flashlight	textbook	thought

Made of Matter

4. _____

5. _____

6. _____

7. _____

8. _____

9. _____

Not Made of Matter

10. _____

11. _____

12. _____

13. _____

14. _____

15. _____

For each statement below, write *true* or *false*.

_____ 16. The mass of an object can vary with the object's location.

_____ 17. A mass measurement includes the effect of Earth's gravitational pull on the object being measured.

_____ 18. Scientists measure the amount of matter in terms of mass.

_____ 19. Subtle differences in weight exist at different locations on Earth.

_____ 20. Your mass on the Moon would be smaller than your mass on Earth.

CHAPTER**1****STUDY GUIDE FOR CONTENT MASTERY****Section 1.2** *continued*

Identify each branch of chemistry described.

21. The study of the matter and processes of living things

22. The study of carbon-containing chemicals

23. The study of the components and composition of substances

24. The study of matter that does not contain organic chemicals

25. The study of the behavior and changes of matter and the related energy changes

For each branch of chemistry in Column A, write the letter of the item in Column B that pertains to that branch.

Column A

_____ **26.** Organic chemistry

_____ **27.** Physical chemistry

_____ **28.** Biochemistry

_____ **29.** Analytical chemistry

_____ **30.** Inorganic chemistry

Column B

a. reaction mechanisms

b. minerals

c. plastics

d. metabolism

e. quality control

Answer the following questions.

31. Compare the macroscopic world with the submicroscopic world.

32. Why are chemists interested in the submicroscopic description of matter?
